

Can Quantum Chemistry Referee Metabolic $\Delta_r G'^{\circ}$?

A QM/ML composite vs. dGPredictor on TECRDB disagreements

ModelSEED thermodynamics project

Progress summary — August 2026

The question, and the short answer

Motivating question

ModelSEED's retrained **dGPredictor** disagrees with experiment (TECRDB) on a set of reactions by 16–100 kJ/mol. Can a **parameter-free quantum-mechanical** method *adjudicate* those disagreements — from physics, sharing none of dGPredictor's group-contribution basis?

Short answer: no — and it need not.

- The data-driven **eQuilibrator** already matches experiment to within its noise (MAE **3.6** kJ/mol, exp. sd 6.2).
- Our QM composite sits at **38.3** kJ/mol — it cannot referee a 100 kJ/mol gap when its own error is ~ 38 .

What QM *can* defensibly do

- On glutathione redox its *magnitude* (~ 36 kJ/mol) independently excludes dGPredictor's ~ 103 —
- from a method trained on **zero** thermodynamic data.
- That corroboration, not a competitive MAE, is the usable result.

Methodology

The composite — what is computed

For every metabolite we build an aqueous standard Gibbs energy, then combine species into reactions and transform to physiological conditions.

$$G_{\text{aq}} = \underbrace{E_{\text{MLIP}}^{\text{gas}}}_{\text{UMA / MACE-POLAR}} + \underbrace{\Delta G_{\text{solv}}^{\text{ALPB}}}_{\text{xtb GFN2}} + \underbrace{G_{\text{RRHO}}}_{\text{quasi-RRHO thermal}}$$

1. Enumerate conformers; optimise **in ALPB water**.
2. Single-point gas energy on that same geometry (electronic + solvation share one structure).
3. One Hessian per compound → thermal correction.
4. **Boltzmann-average** over conformers.
5. Combine into $\Delta_r G'^{\circ}$ per reaction.
6. **Alberty** Legendre transform to the measured pH + extended **Debye–Hückel** (ionic strength).

Key discipline

Corrections are calibrated *only* on external experimental **pKa** values — 2/15

Why this design, and what it is *not*

The error budget (per reaction)

A solvated polyanion computed absolutely carries continuum-solvation error on ions (~ 15 kJ/mol/species), plus conformational, RRHO and electronic error. Four species, uncorrelated:

$$\sqrt{4} \times 15 \approx 30 \text{ kJ/mol}$$

Measured scatter: 42 (130 rxns), 34 (cleanest subset). The gap to the 3–8 kJ/mol “useful” target is **structural**, not a tuning problem — and reproduces Jinich *et al.*, *Sci. Rep.* 2014.

The bottleneck is *not* the electronic method

- Substituting real DFT (r2SCAN-3c) for the MLIP at fixed geometry made it **worse** (14.9 $\rightarrow$ 17.3).
- UMA vs. MACE-POLAR agree at $r = 0.98$ (spread 7 vs. error 40).
- Solvation is the driver: continuum models **under-solvate multiply-charged anions** (~ 30 kJ/mol per reaction).

$\Rightarrow$ every phosphate/polyanion pays the same structural tax.

Implementation

Pipeline — data in, $\Delta_r G^\circ$ out

Nothing is hard-coded to a reaction set: reactions, metabolites and measurement conditions are all read from data.

#	stage	where	what
1	build_inputs	CPU	ModelSEED reactions/metabolites → JSON
2	microspecies	CPU	audit SMILES for protonation / hydration state
3–4	build_ensembles	CPU (xtb)	ETKDG embed → GFN2 opt (ALPB) → Hessian
5	run_uma_ensemble	multi-GPU	UMA gas energy per conformer, Boltzmann avg.
6	analyze_pka	CPU+GPU	anion-solvation calibration (one-time, reusable)
7	final_model	CPU	score reactions → perreaction_dG.csv
8	plot_comparison	CPU	figures

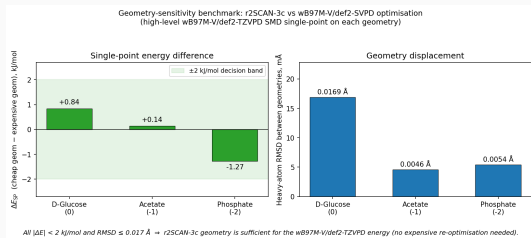
Compute (23 metabolites, 605 conformers):
conformer ensembles ~35 min (dominates, ~2
core-h/metabolite); UMA scoring ~3.5 min on
8×V100.

Scales linearly: ~1000 metabolites ≈ 2000
core-hours ≈ 25 h on 80 cores; GPU time
negligible.

Two engineering checks that matter

Geometry sufficiency

r2SCAN-3c geometry vs. expensive wB97M-V/def2-TZVPD single point: all $|\Delta E| < 2 \text{ kJ/mol}$ and $\text{RMSD} \leq 0.017 \text{ \AA}$.
Cheap geometries are good enough — no expensive re-optimisation.



The Alberty transform passes independently

- ATP hydrolysis -35.3 vs. published -32.5 .
- Debye-Hückel scales as $\Delta(z^2)$ exactly 1:4:9:16.
- pH slope exact.

Bugs found & fixed (each moved the answer):

- Missing standard-state term
 $RT \ln(24.46) = 7.93 \text{ kJ/mol/species}$ (hid in $\Delta n=0$ isomerisations).
- Quality screen rejected acetic acid — now screens on imaginary-mode *magnitude*

Results

Headline — the ten disagreement reactions

method	MAE vs. TECRDB
eQuilibrator	3.6 (exp. sd 6.2)
predict zero	10.5
QM, isodesmic (best)	24.6
GroupContribution	35.4
QM composite	38.3
dGPredictor	61.2

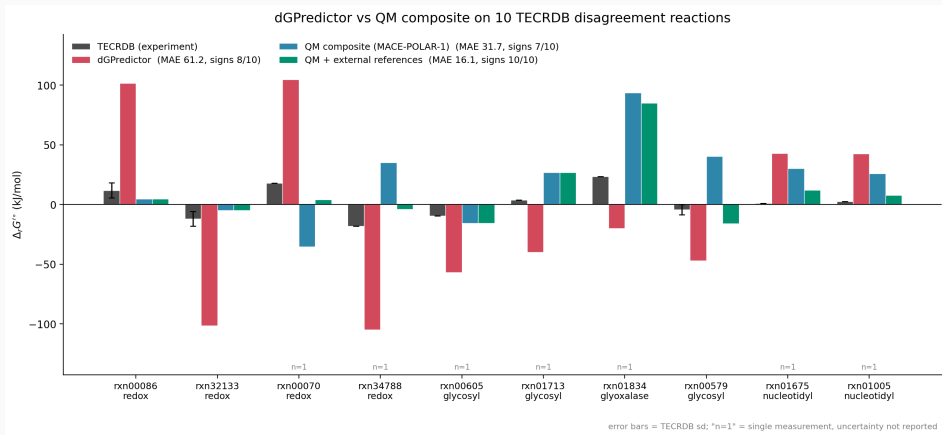
The ten are already solved

eQuilibrator agrees with TECRDB on *exactly* the reactions dGPredictor fails. So the adjudication verdict is “**dGPredictor is the outlier**” — and it needs no quantum chemistry.

Database-wide (1550 TECRDB $\leftrightarrow$ ModelSEED matched reactions): eQuilibrator covers 84% at MAE 5.5; dGPredictor covers 100% at 13.2.

Caveat: the 10 rows are 2 redox chemistries written both directions + 6 single-measurement reactions $\approx$ 8 independent points.

Per-reaction $\Delta_r G'^{\circ}$ — the ten selected reactions



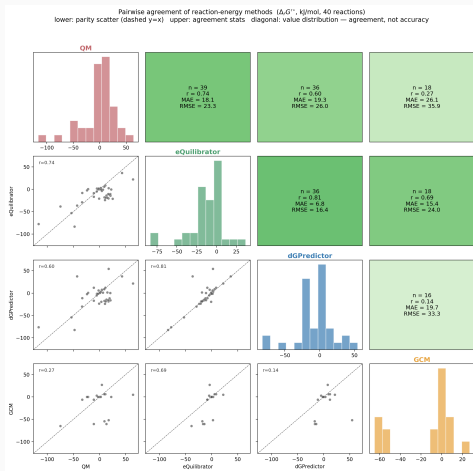
dGPredictor (red) swings to ± 100 kJ/mol on the redox reactions; QM (blue) is closer but wrong-signed on several. Neither tracks the small experimental values (grey).

Per-reaction $\Delta_r G'^{\circ}$ — the numbers

#	reaction	class	TECRDB	dGP	QM
1	rxn00086 glutathione:NADP ⁺ oxidoreductase	redox	11.9	101.6	-25.3
2	rxn32133 glutathione reductase (NADPH)	redox	-11.9	-101.6	25.3
3	rxn00070 glutathione:NAD ⁺ oxidoreductase	redox	18.0	104.9	-46.5
4	rxn34788 glutathione-disulfide reductase (chlpl.)	redox	-18.0	-104.9	46.5
5	rxn00605 UDP-glucose:G6P glucosyltransferase	glycosyl	-9.5	-56.8	-14.3
6	rxn01713 UDP-glucose:sinapate glucosyltransferase	glycosyl	3.9	-39.8	14.2
7	rxn01834 (R)-S-lactoylglutathione lyase	glyoxalase	23.5	-19.8	112.6
8	rxn00579 UDP-glucose:fructose glucosyltransferase	glycosyl	-4.2	-46.9	30.8
9	rxn01675 dTTP:glucose-1-P thymidyltransferase	nucleotidyl	1.0	42.9	39.9
10	rxn01005 UTP:phospho-glucuronate uridylyltransferase	nucleotidyl	2.7	42.7	1.2
MAE vs. TECRDB (kJ/mol)			—	61.2	38.3

All values $\Delta_r G'^{\circ}$ in kJ/mol, pH-matched. dGPredictor's ± 100 kJ/mol redox swing (rows 1–4) is the failure QM was asked to referee; QM's magnitude (~ 36) sits with experiment (15), not with the ± 100 group-contribution regime — the corroboration that matters.

Where the four methods agree, and where they don't



Pairwise agreement over 40 reactions (agreement, *not* accuracy):

- eQuilibrator ↔ dGPredictor agree best ($r = 0.81$, MAE 6.8) — same data lineage.
- QM correlates with all three ($r = 0.6$ – 0.74) but with a wide MAE (~ 18 – 26) — an *independent* axis.
- GCM (GroupContribution) is the weakest ($r \leq 0.27$ vs. QM), and it is what dGPredictor shares a basis with.

The decisive negative result — why no correction rescues it

Per-species anchoring is dead

Compound-grouped cross-validation on per-species offsets:

$$R^2 = 0.177 \quad (\text{in-sample } 0.769)$$

To tie predict-zero needs $R^2 \approx 0.89$; to match eQuilibrator ≈ 0.97 . This rules out every anchoring scheme at once (μ_H , cofactor E'° , anion-site corrections, reference-compound reformulation). The error is not in that subspace.

Mechanisms eliminated

mechanism	verdict
electronic structure	✗ (DFT worse)
MLIP architecture	✗ ($r = 0.98$)
conformer sampling	✗ (wrong sign)
Alberty transform	✓ passes
bond-graph artifact	✗ (0 changed)
protonation state	✓ correct
Mg ²⁺ speciation	✗ (+0.3 kJ/mol net)

Diagnosis is **solid**; every candidate correction was **rejected** on its own test.

A learned component-contribution model

“dGPredictor done right” — a GNN, and an honest test

The idea

Replace the linear group-contribution *lookup table* with a message-passing **GNN** that maps each compound's graph to a scalar formation energy f ; a reaction is the stoichiometric difference

$$\Delta_r G'^{\circ} = \mathbf{S} f, \quad f = \text{GNN}(\text{compound}).$$

Trained on reaction $\Delta_r G'^{\circ}$ *only* (no per-compound labels). **Sum readout** (energy is extensive) + **QC injected as features** (xtb Mulliken charges, HOMO/LUMO, CPCM-X solvation) + **Δ -learning** on the group prior

Why it might beat the incumbents

- No *coverage cliff*: a GNN interpolates in representation space instead of failing on a missing group.
- Nonlinear group interactions + QC physics the linear table cannot see.

The discipline that matters

Every number below is **held-out** cross-validation. The incumbents' famous 3.0 is **in-sample** — they were trained on these reactions.

Result: the model class is *not* the bottleneck

model	eval	rand. CV	cpd.- disj.
eQuilibrator / dGP	<i>in-sample</i>	3.0	—
predict-zero	—	9.7	10.4
linear group-CC	held-out	6.6	8.7
GNN (graph only)	held-out	7.0	8.9
GNN + QC + Δ	held-out	6.8	8.6

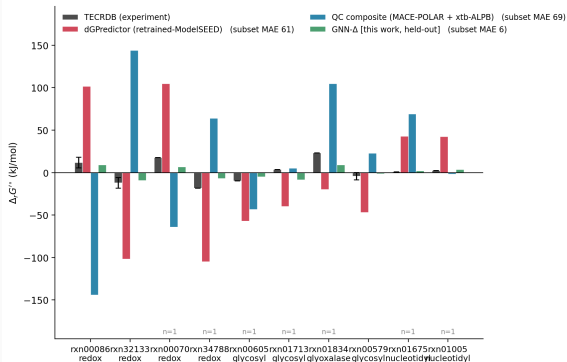
$n = 367$ reactions, 453 compounds. Compound-disjoint = held-out *compounds* (the extrapolation / coverage regime).

- Linear CC, GNN, GNN+QC and GNN+ Δ all land within 0.3 kJ/mol of each other.
- Random held-out $\sim 6.7 \approx$ the **experimental noise floor** (TECRDB sd ~ 6). Little room for *any* model.
- Drive regularization down and in-sample hits 1.6 while held-out worsens — the overfitting made visible.

Verdict

At $n = 367$ the bottleneck is **data, not architecture**. A learned representation buys nothing over regularized group

But it is *robust* — no blow-ups



Cherry-picked disagreement subset (NOT representative). Held-out full-367 MAE: GNN-Δ 6.8 (this work), retrained-dGP 5.7 / standard-dGP 3.0 / eQuilibrator 3.0 (all IN-SAMPLE), raw QC 36. GNN-Δ avoids both the retrained-dGP overfitting blowups and the QC solvation blowups, error bars = TECRDB sd; n=1 = single measurement.

On the cherry-picked hard subset the **GNN-Δ** (green, held-out) tracks experiment — subset MAE **6** — while the retrained-dGP (red, overfit blow-ups) and absolute QC (blue, anion-solvation blow-ups) 13/15

How QC can — and cannot — help

Static QC features: marginal.

Per-compound charges/orbitals are redundant with group counts for the *additive* part of formation energy; $n=367$ is too little to learn the nonadditive part they enable.

QC as a charge-gated prior: physics is right. QC's absolute error $\sim$ Born (z^2) anion under-solvation. A correction in $\Delta(\sum \nu z^2)$ **improves QC-alone** (9.4 $\rightarrow$ 9.1) and *transfers across compounds* — unlike group features.

The catch

On TECRDB, QC still cannot beat group-additivity (9.1 vs 6.7) and adds nothing on top of it (6.94 vs 7.01). Every TECRDB reaction is group-coverable, so QC's real advantage is **invisible here**.

QC's payoff is coverage — the $\sim 42\%$ of ModelSEED with no group decomposition — which has no experimental labels by construction.

$\Rightarrow$ the lever is **training signal**: distil eEquilibrator over **56k** ModelSEED reactions into the GNN (in progress).

Where this leaves us

Summary & where QM should be pointed

What we established

- eQuilibrator solves the TECRDB disagreements; QM cannot beat it on aqueous polyanion chemistry.
- The limit is **continuum anion solvation** (~ 30 kJ/mol/reaction) — structural, reproduces the literature.
- Robust across electronic method, MLIP, CV, solvation swap and Mg^{2+} bookkeeping.
- **The one real QM win:** on glutathione redox its magnitude independently excludes dGPredictor's ± 100 from zero training data

The open problem

- QM is uniquely positioned where data-driven methods *cannot* score: the 16% of matched reactions eQuilibrator misses, stereochemistry (GCM is blind by construction), novel metabolites.
- But that regime has **no validation data by definition**.
- Next: design held-out compound classes / stereoisomer pairs *before* more compute; long-term ceiling is explicit-solvent MD + MLIP + alchemical free energy.

The physics cannot reach 5 kJ/mol on polyanions —
but on the reactions that matter, it agrees with experiment
for reasons the data-driven methods cannot claim.